2022 Alibaba Global Mathematics Competition (Qualifying Round)

1 (Single-Choice Problem) Magic Magnetic Cube

Divide a solid cube $ABCD - A_1B_1C_1D_1$ (with $AB = 1$) into 12 pieces (Figure 1) as follows:

1) Take 6 diagonals of its surfaces AC,AB_1, AD_1, C_1B, C_1D, C_1A_1;

2) Consider all triangles with the center of the cube as a vertex, and one of the above 6 diagonals and 12 edges as the opposite side;

3) These 18 triangles cut the cube into 12 tetrahedra, and each tetrahedron has two edges that are *cube edges*;

4) Each tetrahedron is connected to other tetrahedra **only** by its two cube edges.

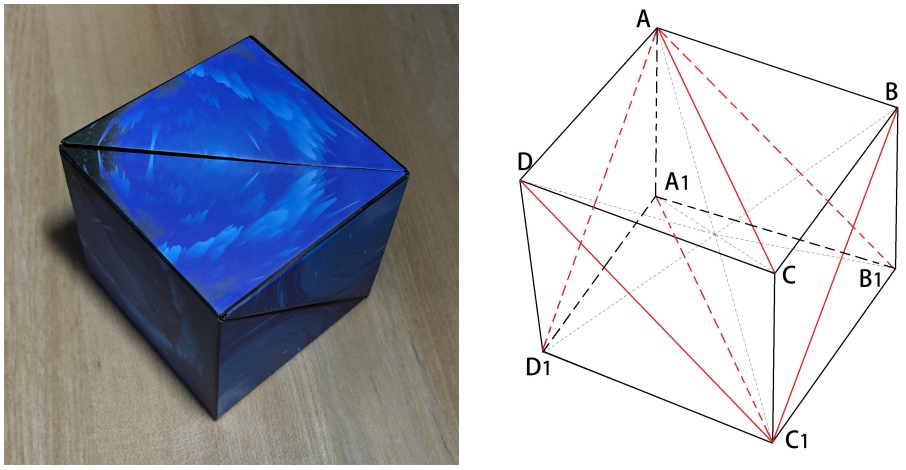


Figure 1: Magic magnetic cube

Such a toy can take on a variety of shapes (Figure 2).

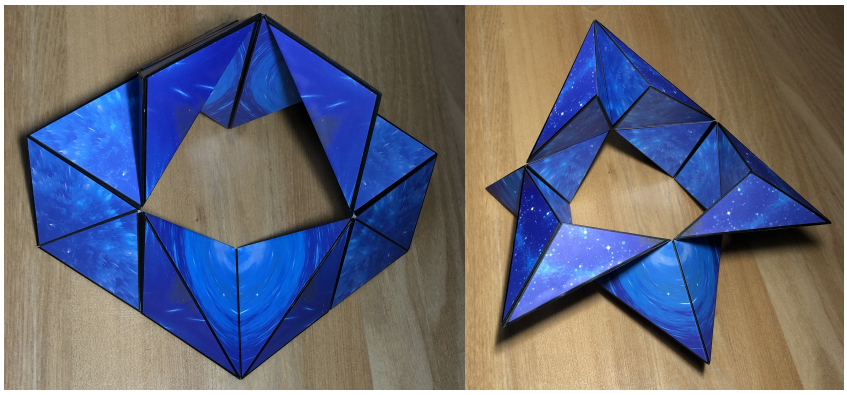


Figure 2: Examples of various shapes

Question: Of all the possible shapes of this toy, what is the maximum distance (in space) between two points on it?

(A) $\sqrt{11}$ (B) $\sqrt{7+4\sqrt{2}}$ (C) $\sqrt{13}$ (D) $1+2\sqrt{2}$ (E) None of the above

1 Answer The answer is **D**. Label the eight vertices of the cube with $ABCD-A_1B_1C_1D_1$, and let the pair of vertices selected at the beginning be A and C_1. Then, these two vertices belong to 6 small tetrahedrons, and B, C, D, D_1, A_1, B_1 (no matter how they are deformed) constitute a Hexagon with side length 1 (possibly degenerate).

Note that this toy consists of 12 tetrahedra (and thus convex), so the maximum value must be the vertices of the two small tetrahedra.

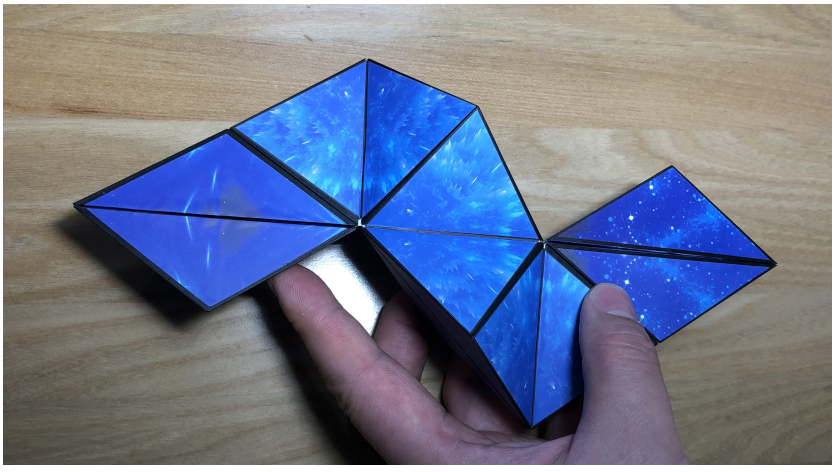
In a tetrahedron, the four vertices can be divided into 3 categories:

- A vertex corresponds to A or C_1 in the cube, we call it a vertex of **Type I**;
- The two vertices (and thus the edges between them) belong to the (spatial) hexagon $BCDD_1A_1B_1$, which we call vertices of **Type II**;
- A vertex corresponds to the center of the cube, and its distance to the other three vertices of the same small tetrahedron is $\frac{\sqrt{3}}{2}$, we call it a vertex of **Type III**.

We examine the maximum value of the distance **along the edges** between these vertices, and it is easy to get the following table:

distance	vertices of Type I	vertices of Type II	vertices of Type III
first kind	$1+2\sqrt{2}$	$2+\sqrt{2}$	$1+\sqrt{2}+\frac{\sqrt{3}}{2}$
second kind	$2+\sqrt{2}$	3	$2+\frac{\sqrt{3}}{2}$
third kind	$1+\sqrt{2}+\frac{\sqrt{3}}{2}$	$2+\frac{\sqrt{3}}{2}$	$2+\sqrt{3}$

Therefore, the distance between any two points on this toy will not exceed $1+2\sqrt{2}$. And this value can be achieved, as shown in the following figure:



2,3 (Single-Choice Problem) Onlook With Distance

One day, there is a Street Art Show at somewhere, and there are some spectators around. We consider this place as an Euclidean plane. Let K be the center of the show. And name the spectators by $A_1, A_2, \ldots, A_n, \ldots$. They pick their positions

$$P_1, P_2, \ldots, P_n, \ldots,$$

one by one. The positions need to satisfy the following three conditions simultaneously.

(i) The distance between K and A_n is no less than 10 meters, that is, $KP_n \geq 10\text{m}$ holds for any positive integer n.

(ii) The distance between A_n and any previous spectator is no less than 1 meter, that is, $P_mP_n \geq 1\text{m}$ holds for any $n \geq 2$ and any $1 \leq m \leq n-1$.

(iii) A_n always choose the position closest to K that satisfies (i) and (ii), that is, KP_n reaches its minimum possible value. If there are more than one point that satisfy (i) and (ii) and have the minimum distance to K, A_n may choose any one of them.

For example, A_1 is not restricted by (ii), so he may choose any point on the circle C which is centered at K with radius 10 meters. For A_2, since there are lots of points on C which are at least 1 meter apart from P_1, he may choose anyone of them.

(1) Which of the following statement is true?

(A) There exist positive real numbers c_1, c_2 such that for any positive integer n, no matter how $A_1, A_2, \ldots, A_n$ choose their positions, $c_1 \leq KP_n \leq c_2$ always hold (unit: meter);

(B) There exist positive real numbers c_1, c_2 such that for any positive integer n, no matter how $A_1, A_2, \ldots, A_n$ choose their positions, $c_1\sqrt{n} \leq KP_n \leq c_2\sqrt{n}$ always hold (unit: meter);

(C) There exist positive real numbers c_1, c_2 such that for any positive integer n, no matter how $A_1, A_2, \ldots, A_n$ choose their positions, $c_1 n \leq KP_n \leq c_2 n$ always hold (unit: meter);

(D) There exist positive real numbers c_1, c_2 such that for any positive integer n, no matter how $A_1, A_2, \ldots, A_n$ choose their positions, $c_1 n^2 \leq KP_n \leq c_2 n^2$ always hold (unit: meter).

(2) Since human bodies are 3-dimensional, if one spectator's position is near another spectator's path of view, then the second one's sight will be blocked by the first one. Suppose that for different i, j, if the circle centered at P_i with radius $\frac{1}{6}$ meter intersects with segment KP_j, then A_j's sight will be blocked by A_i, and A_j could not see the entire show.

Which of the following statement is true?

(A) If there were 60 spectators, then some of them could not see the entire show;

(B) If there were 60 spectators, then it is possible that all spectators could see the entire show, but if there were 800 spectators, then some of them could not see the entire show;

(C) If there were 800 spectators, then it is possible that all spectators could see the entire show, but if there were 10000 spectators, then some of them could not see the entire show;

(D) If there were 10000 spectators, then it is possible that all spectators could see the entire show.

2 Answer The answer is **B**. Suppose the length of KP_n is d_n meters.

We consider the discs centered at $P_1, P_2, \ldots, P_{n-1}$ with radius 1 meter. Use the property of P_n we get that these discs and the interior of C cover the disc centered at K with radius d_n, so

$$\pi \cdot d_n^2 \leq (n-1) \cdot \pi \cdot 1^2 + \pi \cdot 10^2,$$

It follows that

$$d_n \leq \sqrt{n+99} \leq \sqrt{100n} = 10\sqrt{n}.$$

On the other hand, we consider the discs centered at $P_1, P_2, \ldots, P_n$ with radius $\frac{1}{2}$ meter. Since the distance between any two of $P_1, P_2, \ldots, P_n$ is not less than 1 meter, all these discs do not intersect.

Note that every length of $KP_1, KP_2, \ldots, KP_n$ is not more than d_n meter (If the length of KP_m is more than d_n meter, then A_m can choose P_n, which is closer to K, contradiction.)

So all these discs are inside the circle centered at K with radius $d_n + \frac{1}{2}$, and

$$\pi(d_n + \frac{1}{2})^2 \geq n \cdot \pi \cdot (\frac{1}{2})^2,$$

So

$$d_n \geq \frac{\sqrt{n}}{2} - \frac{1}{2}.$$

For $n = 1$, $d_1 = 10$; For $n \geq 2$, we have that $\frac{1}{2} < \frac{2\sqrt{n}}{5}$, so

$$d_n > \frac{\sqrt{n}}{2} - \frac{2\sqrt{n}}{5} = \frac{\sqrt{n}}{10}.$$

Therefore, $\frac{\sqrt{n}}{10} \leq d_n \leq 10\sqrt{n}$, (B) is correct.

3 Answer The answer is **B**. For 60 residents, since $\sin\frac{\pi}{60} > \frac{1}{10}\sin\frac{\pi}{6} = \frac{1}{20}$, the side length of a regular 60-gon inscribed in C is not less than 1 meter. Therefore, $P_1, P_2, \ldots, P_{60}$ may be all vertices of this polygon. For different i, j, the distance from P_i to KP_j is not less than $10\sin\frac{\pi}{30}$ meter. Since $\sin\frac{\pi}{30} > \frac{1}{5}\sin\frac{\pi}{6} = \frac{1}{10}$, all residents could see the entire show.

On the other hand, if there are 800 residents, we draw rays $\overrightarrow{KP_1}, \overrightarrow{KP_2}, \ldots, \overrightarrow{KP_{800}}$. If two of them (called $\overrightarrow{KP_i}, \overrightarrow{KP_j}$) coincide, and $KP_j > KP_i$, then A_j's sight line is blocked by A_i.

Suppose no two rays coincide, we first prove that if an angle $\angle P_iKP_j$ satisfies $\angle P_iKP_j \le \frac{1}{12}\left(\frac{1}{KP_i}+\frac{1}{KP_j}\right)$ (the unit of angle is rad, and the unit of length is meter), then the sight line of one of A_i, A_j is blocked by the other.

Without loss of generality we suppose $KP_i \le KP_j$. Since $KP_i, KP_j \ge 10$, we get that $\angle P_iKP_j \le \frac{1}{60}$, so it is acute. Therefore, the foot point from P_i to KP_j is inside segment KP_j, and the distance from P_i to KP_j is

$$KP_i \sin\angle P_iKP_j < KP_i \cdot \angle P_iKP_j \le \frac{KP_i}{12}\Big(\frac{1}{KP_i}+\frac{1}{KP_j}\Big) \le \frac{1}{6},$$

So, A_j's sight line is blocked by A_i.

Note that $\overrightarrow{KP_1}, \overrightarrow{KP_2}, \ldots, \overrightarrow{KP_{800}}$ cut the perigon with vertex K to 800 angles, and their sum is 2π, but the sum of all $\frac{1}{12}\left(\frac{1}{KP_i}+\frac{1}{KP_j}\right)$ is

$$\begin{aligned}
\sum \frac{1}{12}\Big(\frac{1}{KP_i}+\frac{1}{KP_j}\Big) &= \frac{1}{6}\sum_{i=1}^{800}\frac{1}{KP_i} \\
&\ge \frac{1}{6}\sum_{i=1}^{800}\frac{1}{\sqrt{i+99}} \quad \text{(from the conclusion of (1))} \\
&= \frac{1}{6}\sum_{m=100}^{899}\frac{1}{\sqrt{m}} \\
&\ge \frac{1}{6}\sum_{m=100}^{899}\Big(\int_m^{m+1}\frac{1}{\sqrt{x}}dx\Big) \quad (\text{since}\frac{1}{\sqrt{x}}\text{is a decreasing function}) \\
&= \frac{1}{6}\int_{100}^{900}\frac{1}{\sqrt{x}}dx = \frac{\sqrt{900}-\sqrt{100}}{3} = \frac{20}{3} > 2\pi.
\end{aligned}$$

Therefore, there exist an angle $\angle P_iKP_j$ satisfies $\angle P_iKP_j \le \frac{1}{12}\left(\frac{1}{KP_i}+\frac{1}{KP_j}\right)$, and some of the residents could not see the entire show.

4,5 (Single-Choice Problem) Tiger Mystery Box

A milk drink company organizes a promotion during the Chinese New Year:
one gets a red packet when buying a carton of milk of their brand, and there is one of the following characters in the red packet " 虎"(Tiger), "生"(Gain), "威"(Strength).

If one collects two "虎", one "生" and one "威", then they form a Chinese phrases "虎虎生威"(Pronunciation: hu hu sheng wei), which means "Have the courage and strength of the tiger". This is a nice blessing because the Chinese zodiac sign for the year 2022 is tiger. Soon, the product of Brave NiuNiu becomes quite popular and people hope to get a collection of "虎虎生威". Suppose that the characters in every packet are independently random, and each character has probability $\frac{1}{3}$.

(1) What is the expectation of cartons of milk to collect "虎虎生威"(i.e. one collects at least 2 copies of "虎", 1 copy of "生", 1 copy of "威")?

(A) $6\frac{1}{3}$

(B) $7\frac{1}{3}$

(C) $8\frac{1}{3}$

(D) $9\frac{1}{3}$

(E) None of the above

(2) In a weekly meeting of Brave NiuNiu, its market team notices that one often has to collect too many "生" and "威", before getting a collection of "虎虎生威". Thus an improved plan is needed for the proportion of characters. Suppose that the probability distribution of "虎", "生" and "威" is (p, q, r), then which of the following plans has the smallest expectation (among the 4) for a collection of "虎虎生威"?

(A) $(p, q, r) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$

(B) $(p, q, r) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$

(C) $(p, q, r) = (\frac{2}{5}, \frac{3}{10}, \frac{3}{10})$

(D) $(p, q, r) = (\frac{3}{4}, \frac{1}{8}, \frac{1}{8})$

4 Answer The answer is **B**. We can use Poisson process to get the explicit formula for the general case. Suppose that there are in total n characters. The probability for the character i is p_i, and we aim to collect k_i copies of the character i. We denote by N the first time to realize our collection, and we need to calculate $\mathbb{E}[N]$. A nice technique is embedding this model to a Poisson process: we have a Poisson process of density 1. Every time when the signal arrives, we sample independently p_i for the character i. We also denote by

$$\begin{aligned} T_i &= \inf\{t \in \mathbb{R}_+ : \text{ before } t \text{ one collects } k_i \text{ copies of the character } i\}, \\ T &= \max_{1 \le i \le n} T_i. \end{aligned}$$

We claim that $\mathbb{E}[T] = \mathbb{E}[N]$. Let us prove this claim. We denote by τ_j the waiting time for the j-th signal, then we have

$$T = \sum_{j=1}^{N} \tau_j.$$

By the property of Poisson process, $(\tau_j)_{j \ge 1}$ are i.i.d. exponential random variable. Using conditional probability we have

$$\begin{aligned} \mathbb{E}[T] &= \mathbb{E}[\sum_{j=1}^{N} \tau_j] \\ &= \sum_{k=0}^{\infty} \mathbb{E}\left[\sum_{j=1}^{k} \tau_j | N = k\right] \mathbb{P}[N = k] \\ &= \sum_{k=0}^{\infty} \mathbb{P}[N = k] k \mathbb{E}[\tau_1] \\ &= \mathbb{E}[N]. \end{aligned}$$

This justifies our claim.

Then it suffices to calculate $\mathbb{E}[T]$. By Fubini's lemma

$$\begin{aligned} \mathbb{E}[T] &= \int_0^{\infty} \mathbb{P}[T > t] \, dt \\ &= \int_0^{\infty} (1 - \mathbb{P}[T \le t]) \, dt \\ &= \int_0^{\infty} (1 - \mathbb{P}[T_i \le t, \forall 1 \le i \le n]) \, dt. \end{aligned}$$

In fact, the thinning property of the Poisson process creates a lot of independence. We can treat the collection of the character i as independent Poisson processes of parameter p_i. Then we have

$$\mathbb{E}[T] = \int_0^{\infty} \left(1 - \prod_{i=1}^{n} \mathbb{P}[T_i \le t]\right) dt.$$

We write down directly the explicit formula of $\mathbb{P}[T_i \le t]$ using Poisson distribution

$$\mathbb{P}[T_i \le t] = 1 - \sum_{k=0}^{k_i-1} e^{-p_i t} \frac{(p_i t)^k}{k!}.$$

Finally, it gives us

$$\mathbb{E}[N] = \int_0^\infty \left(1 - \prod_{i=1}^{n} \left(1 - \sum_{k=0}^{k_i-1} e^{-p_i t} \frac{(p_i t)^k}{k!}\right)\right) dt.$$

In our setting, $n = 3$, and the object $(k_1, k_2, k_3) = (2, 1, 1)$. Thus we have

$$\mathbb{E}[N] = 1 + p_1 + \left(\frac{2}{p_1} + \frac{1}{p_2} + \frac{1}{p_3}\right) - \sum_{i=1}^{3} \frac{1}{1-p_i} - \frac{p_1}{(p_1+p_2)^2} - \frac{p_1}{(p_1+p_3)^2}.$$

When $(p_1, p_2, p_3) = (1/3, 1/3, 1/3)$, the expectation is $7\frac{1}{3}$.

5 Answer The answer is **C**. In last question, we know the expectation for Plan A is $7\frac{1}{3}$. Plan D is not a good plan obviously, because the expectation to collect "威" is 8, which is larger than Plan A. It suffices to calculate Plan B and C. Using the expression

$$\mathbb{E}[\tau] = 1 + p + \left(\frac{2}{p} + \frac{1}{q} + \frac{1}{r}\right) - \left(\frac{1}{p+q} + \frac{1}{p+r} + \frac{1}{q+r}\right) - \frac{p}{(p+q)^2} - \frac{p}{(p+r)^2}.$$

The expectation for Plan B and Plan C are respectively $7\frac{1}{18}, 6\frac{223}{245}$. Plan C is the best one.

6 (Proof Question)

Given a set X and a function $f : X \times X \to [0, 1]$, we say f is right uniform if for any $\epsilon > 0$, there exist finitely many elements $b_1, b_2, ..., b_m$ in X such that for any $t \in X$, the following holds for some $b_{i(t)}$:

$$|f(x, t) - f(x, b_{i(t)})| < \epsilon, \forall x \in X.$$

Similarly, we say f is left uniform if for any $\epsilon > 0$, there exist finitely many elements $a_1, a_2, ..., a_n$ in X such that for any $t \in X$, the following holds for some $a_{i(t)}$:

$$|f(t, x) - f(a_{i(t)}, x)| < \epsilon, \forall x \in X.$$

Prove that f is right uniform if and only if it's left uniform.

6 Answer It suffices to prove that if f is right uniform, then it's left uniform. Given $\epsilon > 0$, we have finitely many elements $b_1, b_2, ..., b_m$ as in the definition of being right uniform. Consider the map $h : X \to [0, 1]^m, h(x) = (f(x, b_1), f(x, b_2), ..., f(x, b_m))$. Because $[0, 1]^m$ is compact, there are $c_1, c_2, ..., c_n$ in X such that for any $x \in X$, $|h(x) - h(c_{i_x})| < \epsilon$ holds for some c_{i_x}. Hence

$$|f(x, b_i) - f(c_{i_x}, b_i)| < \epsilon, \forall i \in \{1, 2, ..., m\}. \tag{1}$$

Now since f is right uniform, we have $|f(x, t) - f(x, b_{i_t})| < \epsilon (\forall x \in X), |f(c_{i_x}, t) - f(c_{i_x}, b_{i_t})| < \epsilon$. Combined with (1), it follows that for any $t \in X$,

$$|f(x, t) - f(c_{i_x}, t)| \leq |f(x, t) - f(x, b_{i_t})| + |f(x, b_{i_t}) - f(c_{i_x}, b_{i_i})| + |f(c_{i_x}, b_{i_t}) - f(c_{i_x}, t)| < 3\epsilon.$$

Since ϵ is arbitrary, f is left uniform.

7 (Proof Questions)

Let n be a positive integer and $V = \mathbb{R}^n$ be an n-dimensional Euclidean space with a basis $e_i = (\underbrace{0, \ldots, 0}_{i-1}, 1, \underbrace{0, \ldots, 0}_{n-i})$ $(1 \leq i \leq n)$ and with an inner product $(\cdot, \cdot)$ defined by

$$(e_i, e_j) = \delta_{i,j}$$

where

$$\delta_{i,j} = \begin{cases} 1 \text{ if } i = j \\ 0 \text{ if } i \neq j \end{cases}$$

is Kronecker's symbol. For a nonzero vector $v \in V$, define $s_v : V \to V$ by

$$s_v(u) = u - \frac{2(u, v)}{(v, v)} v, \ \forall u \in V.$$

For an integer k between 0 and n, write $\mathrm{Gr}_k(V)$ for the set of k-dimensional subspaces of V. For a k-dimensional subspace W of V, write $[W]$ for the corresponding element of $\mathrm{Gr}_k(V)$. Choose an orthonomal basis $\{v_1, \ldots, v_k\}$ of W, define $s_{[W]} : V \to V$ by

$$s_{[W]} = s_{v_1} \cdots s_{v_k}.$$

(1) Prove that $s_{[W]}$ is independent of the choice of an orthonomal basis $\{v_1, \ldots, v_k\}$.

(2) Prove that $s_{[W]}^2 = \mathrm{id}$.

(3) For another element $[W'] \in \mathrm{Gr}_k(V)$, define

$$t_{[W]}([W']) = [s_{[W]}(W')],$$

where $s_{[W]}(W')$ is the image of W' under $s_{[W]}$. We call a subset X of $\mathrm{Gr}_k(V)$ a "nice set" if

$$t_{[W]}([W']) = [W'], \ \forall [W], [W'] \in X.$$

Find the maximal cardinality of a "nice set" in $\mathrm{Gr}_k(V)$ and prove it.

7 Answer

(1) Write $W^\perp$ for the orthogonal complement of W in V. Then,

$$s_{[W]}|_W = 1 \text{ and } s_{[W]}|_{W^\perp} = -1.$$

This characterizes $s_{[W]}$ and shows its independence with the choice of an orthonomal basis of W.

(2) Since

$$s_{[W]}|_W = 1 \text{ and } s_{[W]}|_{W^\perp} = -1,$$

then $s_{[W]}^2 = \mathrm{id}$.

(3) (i)We show that

$$t_{[W]}([W']) = [W'] \Leftrightarrow W' = (W' \cap W) \oplus (W' \cap W^\perp).$$

The sufficiency is clear. For the necessarity, suppose that $t_{[W]}([W']) = [W']$. Then, for any $u \in W'$, we have $s_{[W]}(u) \in W'$. Write $u = u_1 + u_2$ where $u_1 \in W$ and $u_2 \in W^\perp$. Then, $u_1 - u_2 = s_{[W]}(u) \in W'$. Thus, $u_1 \in W'$ and $u_2 \in W'$. Hence,

$$W' \subset (W' \cap W) \oplus (W' \cap W^\perp).$$

It is clear that $(W' \cap W) \oplus (W' \cap W^\perp) \subset W'$. Therefore,

$$W' = (W' \cap W) \oplus (W' \cap W^\perp).$$

(ii) Put

$$X = \{[\mathrm{span}_{\mathbb{R}}\{e_{i_1}, \ldots, e_{i_k}\}] : 1 \le i_1 < \cdots < i_k \le n\}.$$

By the above criterion in (i), X is a nice set in $\mathrm{Gr}_k(V)$. This nice set X has cardinality $\binom{k}{n}$.

(iii) We prove by induction on n that any nice set X in $\mathrm{Gr}_k(V)$ has cardinality at most $\binom{k}{n}$. When $n = 1$, this is trivial. Suppose the conclusion holds when $n < m$. Assume that $n = m$. When $k = 0$ or m, the conclusion is trivial. Assume that $1 \le k \le m-1$. Choose an element $[W] \in X$. For each i $(0 \le i \le k)$, write

$$X_i = \{[W'] \in X : \dim W' \cap W = k - i\}.$$

For each $[W'] \in X_i$, put

$$Y_i = \{[W' \cap W] : [W'] \in X_i\} \subset \mathrm{Gr}_{k-i}(W)$$

and

$$Z_i = \{[W' \cap W^\perp] : [W'] \in X_i\} \subset \mathrm{Gr}_i(W^\perp).$$

By the criterion in (i), Y_i is a nice set in $\mathrm{Gr}_{k-i}(W)$ and Z_i is a nice set in $\mathrm{Gr}_i(W^\perp)$. Since $\dim W = k < m$ and $\dim W^\perp = m - k < m$, by induction we get

$$|Y_i| \le \binom{k-i}{k} \text{ and } |Z_i| \le \binom{i}{m-k}.$$

Therefore,

$$|X| = \sum_{0 \le i \le k} |X_i| \le \sum_{0 \le i \le k} |Y_i||Z_i| \le \sum_{0 \le i \le k} \binom{k-i}{k}\binom{i}{m-k} = \binom{k}{m}.$$

This shows the conclusion.

8 A Busy Courier

A courier picks up a package at coordinate $(n, 0)$ in the two-dimensional lattice and then does a discrete time simple random walk on $\mathbb{Z}^2$. His station locates at the origin $(0, 0)$. In the rest of this question, you may without loss of generality assume n is sufficiently large.

(1) Let $P_{1,n}$ be the probability that at his $\lfloor n^{1.5} \rfloor$th step, the distance between this courier and his station is greater than $\frac{n}{2}$. Prove that

$$\lim_{n \to +\infty} P_{1,n} = 1$$

(2) Let $P_{2,n}$ be the probability that the courier has ever reached the station within his first $\lfloor n^{1.5} \rfloor$ steps. Prove that

$$\lim_{n \to +\infty} P_{2,n} = 0$$

(3) Let $P_{3,n}$ be the probability that the courier has ever reached the station within his first 2^n steps. Prove that

$$\lim_{n \to +\infty} P_{3,n} = 1$$

8 Answer (1) Consider the event that this courier is of a distance no more than $\frac{n}{2}$ from the station after $\lfloor n^{1.5} \rfloor$ steps. By triangular inequality, such event has to be a subset of event A defined as follows:

$$A = \text{the courier is of a distance at least } \frac{n}{2} \text{ from where}$$
$$\text{he picked up the package after } \lfloor n^{1.5} \rfloor \text{ steps}$$

Again by triangular inequality, if event A happens, then the courier has to move at least $\frac{n}{4}$ along at least one coordinate. Without loss of generality, one may assume it is the y-coordinate. Then the displacement of the courier along y-coordinate can be written as the following i.i.d. sum:

$$Y_k = X_1 + X_2 + \cdots + X_k$$

where $P(X_i = 0) = \frac{1}{2}, P(X_i = \pm 1) = \frac{1}{4}$ with $E(X_i) = 0, var(X_i) = \frac{1}{2}$. So by Chebyshev Inequality,

$$P(A) \leq 2P\Big(\Big|Y_{\lfloor n^{1.5} \rfloor}\Big| \geq \frac{n}{4}\Big) \leq 2 \times \frac{n^{1.5}}{2n^2/16} = 16n^{-0.5} << 1$$

(2) Similar to Part (1), by triangular inequality, the event of interest has to be a subset of event A defined as follows:

$$A = \text{the courier } \textbf{has ever reached} \text{ a distance at least } n \text{ from where}$$
$$\text{he picked up the package again } \lfloor n^{1.5} \rfloor \text{ steps}$$

If A happens, then the courier has to have wandered at least $\frac{n}{2}$ along at least one coordinate. I.e.,

$$P(A) \leq 2P\Big(\max_{k \leq n^{1.5}} |Y_k| \geq \frac{n}{2}\Big)$$

Note that Y_k forms a martingale, by Doob's inequality,

$$P\Big(\max_{k \leq n^{1.5}} |Y_k| \geq \frac{n}{2}\Big) \leq \frac{E\Big[\Big|Y^2_{\lfloor n^{1.5} \rfloor}\Big|\Big]}{(n/2)^2} \leq \frac{n^{1.5}}{2(n/2)^2} = 2n^{-0.5} << 1$$

(3) Let $S_k \in \mathbb{Z}^2, k \geq 0$ be the location of courier at kth step. Define stopping times as follows:

$$\tau = \min\{k \geq 0, S_k = 0\}$$
$$\widehat{\tau} = \min\{k \geq 0, |S_k| \geq 2^{n/4}\}$$

Thus for the probability of interest we have,

$$P(\tau > 2^n) \leq P_{(n,0)}(\widehat{\tau} > 2^n) + P_{(n,0)}(\widehat{\tau} < \tau)$$

where $P_{(n,0)}$ stands for the distribution of simple random walk starting from $(n, 0)$. For the first term on RHS of the inequality above, by invariance principle, we have $S_{[2^{n/2}t]}2^{-\frac{n}{4}}$, $t \in [0, 1]$ converges to a $2D$ standard Brownian Motion B_t, where

$$P(|B_1| \geq 2) = c > 0$$

Thus for all sufficiently large n,

$$\min_{|X_0|\le 2^{n/4}} P_{X_0}(|S_{2^{n/2}}| \ge 2^{n/4}) \ge P_0(|S_{2^{n/2}}| \ge 2\times 2^{n/4}) \ge P(|B_1| \ge 2) - \frac{c}{2} \ge \frac{c}{2}$$

Then we partite the 2^n into subintervals of length $2^{n/2}$. If the courier never level the neighborhood of radius $2^{n/4}$ within 2^n steps, he has to be so at time points $2^{n/2}, 2\times 2^{n/2}, \cdots, [2^{n/2}]2^{n/2}$ as wells. Thus by Markov property,

$$P_{(n,0)}(\widehat{\tau} > 2^n) \le \Big[1 - \min_{|X_0|\le 2^{n/4}} P_{X_0}(|S_{2^{n/2}}| \ge 2^{n/4})\Big]^{[2^{n/2}]} \le \Big(1 - \frac{c}{2}\Big)^{[2^{n/2}]} << 1$$

Now for the second term $p_2 = P_{(n,0)}(\widehat{\tau} < \tau)$ we need to consider the potential kernel of $2D$ simple random walk, for any $x \in \mathbb{Z}^2$ define

$$a(x) = \sum_{k=0}^{+\infty} [P_0(S_k = 0) - P_0(S_k = x)]$$

By Kesten (1987, Lemma 1)[1], we have

1) $a(x)$ converges for all $x \in \mathbb{Z}^2$ and is nonnegative
2) For SRW S_k starting from any initial point, $a(S_{k\wedge\tau})$ is a nonnegative martingale
3) There is a constant C_0 such that

$$|a(x) - \log|x|/(2\pi) - C_0| = O(|x|^{-2})$$

Thus by martingale optional stopping theorem, we have

$$a((n,0)) = (1-p_2)\times 0 + p_2 E[a(S_{\widehat{\tau}})|\tau > \widehat{\tau}]$$

Thus

$$p_2 = \frac{a((n,0))}{E[a(S_{\widehat{\tau}})|\tau > \widehat{\tau}]}$$

Recalling Condition 3 above,

$$a((n,0)) \asymp \log(n)$$
$$E[a(S_{\widehat{\tau}})|\tau > \widehat{\tau}] \asymp n$$

Thus we have $p_2 << 1$ and complete the proof.

[1]H. Kesten, Hitting probabilities of random walks on $\mathbb{Z}^d$, Stochastic Process. Appl. 25 (1987) 165-184.

9 Questions

(1) Show that there is no *periodic* sequence $a_1, a_2, a_3, \ldots$ of signs $a_n \in \{\pm 1\}$ such that, for every *rational* number θ,

$$\sup_{N\in\mathbb{N}} \left| \sum_{n=1}^{N} a_n e^{2\pi i n\theta} \right| < +\infty.$$

(2) Show that there is no sequence $a_1, a_2, a_3, \ldots$ of signs $a_n \in \{\pm 1\}$ such that, for every *rational* number θ

$$\sup_{N\in\mathbb{N}} \left| \sum_{n=1}^{N} a_n e^{2\pi i n\theta} \right| \leq 2022\,?$$

(3) Give an example of a sequence of signs $a_n \in \{\pm 1\}$ such that, for every $\theta \in \mathbb{Q} - \mathbb{Z}$,

$$\sup_{N\in\mathbb{N}} \left| \sum_{n=1}^{N} a_n e^{2\pi i n\theta} \right| < +\infty,$$

and which takes each of the values $+1$ and -1 an infinite number of times.

9 Answer **(1)** To simplify the notation, we shall write $z := e^{2\pi i\theta}$. Suppose, by contradiction, that there is a sequence (a_n), periodic with period $d \in \mathbb{N}$, such that

$$\sup_{N\in\mathbb{N}} \left| \sum_{n=1}^{N} a_n z^n \right| < +\infty,$$

whenever z is of the form $e^{2\pi i\theta}$ with $\theta \in \mathbb{Q}$.

For convenience, take N to be a multiple d, say $N = dM$. Since (a_n) is periodic, we can group the terms of the above sum as

$$\begin{aligned}\sum_{n=1}^{Md} a_n z^n =& (a_1 z + a_2 z^2 + \dots a_d z^d) + z^d(a_1 z + a_2 z^2 + \dots a_d z^d) \\ &+ \dots + z^{(M-1)d}(a_1 z + a_2 z^2 + \dots a_d z^d) \\ =& (a_1 z + a_2 z^2 + \dots a_d z^d)\left(1 + z^d + z^{2d} + \dots + z^{(M-1)d}\right).\end{aligned}$$

Take $\theta = k/d \in \mathbb{Q}$, for some $k \in \{0, \dots, d-1\}$ to be chosen later. Then $z^d = \left(e^{2\pi ik/d}\right)^d = 1$, so

$$\sum_{n=1}^{Md} a_n z^n = M(a_1 z + a_2 z^2 + \dots a_d z^d).$$

To conclude the problem, it suffices to show that $a_1 z + a_2 z^2 + \dots a_d z^d$ in nonzero for some appropriate choice of k, since we would then have

$$\lim_{M\to+\infty} \left| \sum_{n=1}^{Md} a_n z^n \right| = \lim_{M\to+\infty} M|a_1 z + a_2 z^2 + \dots a_d z^d| = +\infty,$$

contradicting **??**.

This is clear: if $a_1 z + a_2 z^2 + \dots a_d z^d$ vanished for all $k \in \{0, \dots, d-1\}$ (with $z = e^{2\pi ik/d}$), then the polynomial $P(X) = a_1 X + a_2 X^2 + \dots a_d X^d$ would have $d+1$ distinct complex roots

$$e^{2\pi i 0/d},\ e^{2\pi i 1/d},\ \dots,\ e^{2\pi i(d-1)/d} \quad \text{and} \quad 0,$$

which is impossible for a degree d polynomial.

(2) Suppose, by contradiction, that there is such a sequence (a_n). At first, since

$$f_N := \sum_{n=1}^{N} a_n e^{2\pi in\theta}$$

is a continous function of θ, $|f_N(\theta)| \le 2022$ for any $\theta \in \mathbb{R}$.

Now we calculate the L^2 norm of f_N. Notice that for $n \in \mathbb{Z}$,

$$\int_0^1 e^{2\pi in\theta}\, d\theta = \begin{cases} 1 & \text{if } n = 0, \\ 0 & \text{if } n \neq 0. \end{cases}$$

Then we have

$$\begin{aligned}\int_0^1 |f_N|^2 d\theta &= \int_0^1 \sum_{n=1}^N a_n e^{2\pi i n\theta} \sum_{m=1}^N a_n e^{-2\pi i m\theta} d\theta \\ &= \sum_{n=1}^N \sum_{m=1}^N a_n a_m \underbrace{\int_0^1 e^{2\pi i(n-m)\theta} d\theta}_{\text{nonzero only if } n=m} \\ &= \sum_{n=1}^N a_n^2 = N\end{aligned}$$

This shows that (a_n) cannot have the property that

$$\left| \sum_{n=1}^N a_n e^{2\pi i n\theta} \right| \leq 2022$$

for all N and $\theta \in \mathbb{R}$! If this were true, we would have $|f_N(\theta)| \leq 2022$ for all N and $\theta \in \mathbb{R}$. $\int_0^1 |f_N|^2 d\theta \leq 2022^2$, contradicting $\int_0^1 |f_N|^2 d\theta = N$ if $N > 2022^2$.

(3) (We remark that eventually-constant sequences satisfy the inequality by the geometric series formula, but are explicitely discarded in the statement of the problem.)

To build up some intuition for the problem, let us take $\theta = 1/3$ and $N = 3M$. We want to find a sequence a_i such that

$$\sup_{M\in\mathbb{N}} \left| \left(\sum_{n=0}^{M-1} a_{3n+1} \right) \cdot 1 + \left(\sum_{n=0}^{M-1} a_{3n+2} \right) \cdot e^{2\pi i/3} + \left(\sum_{n=0}^{M-1} a_{3n+3} \right) \cdot e^{4\pi i/3} \right| < +\infty$$

Geometrically, this corresponds to scaling the three vectors 1, $e^{2\pi i/3}$ and $e^{4\pi i/3}$, and computing the norm of the sum of these scaled vectors.

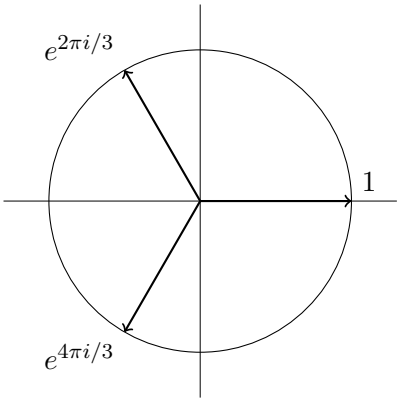


Figure 3: Case $\theta = 1/3$.

Since $1 + e^{2\pi i/3} + e^{4\pi i/3} = 0$, we should try to scale these three vectors by almost the same amount in order to obtain good cancellation. In other words, the distribution of the number of 1's and -1's in the sequence $(a_{3n+i})_{n\geq 1}$ should be approximately the same for $i = 1$, $i = 2$ and $i = 3$. Of course, a similar property should hold for any natural number (greater that 1 since $\theta \notin \mathbb{Z}$) in place of 3.

There is a variety of ways to construct such a sequence (a_n). One way to do it is as follows:

n	1	2	3	4	5	6	7	8	9	10	11
a_n	$+1$	-1	-1	$+1$	$+1$	$+1$	$+1$	$+1$	$+1$	-1	-1

,

where $+1$ appears 1! time, then -1 appears 2! times, then $+1$ appears $3! = 6$ times, and so on. Let us now prove that this sequence has the desired property.

For $k \geq 0$, let $f(k) = 0! + 1! + \cdots + k!$, and let $I_k \subset \mathbb{N}$ be the set of integers n with $f(k) \leq n < f(k+1)$. The sequence above is that which takes the value $a_n = (-1)^k$ for $n \in I_k$. The key to our construction is that, for every integer $q \geq 2$, the length of I_k, namely $(k+1)!$, is divisible by q whenever k is sufficiently large.

Let $\theta = p/q$ be a rational number, with $\gcd(p, q) = 1$ and $q > 1$. Then

$$\sum_{n=1}^{N} a_n e^{2\pi i n\theta} = \sum_{k\geq 0} (-1)^k \sum_{\substack{n\in I_k \\ n\leq N}} e^{2\pi i n p/q}.$$

We need to show that this expression is bounded above by a constant that does not depend on N.

Consider, for each fixed $k \geq 0$, the inner sum $S_k := \sum_{\substack{n\in I_k \\ n\leq N}} e^{2\pi i n p/q}$.

(i) If $k < q$, we will be able to use the trivial bound $|S_k| \leq f(k+1) - f(k) = (k+1)!$ (obtained by the triangle inequality). This will be enough since there are only a finite number of such k.

(ii) If $k \geq q$, we will exploit the particular way in which we constructed the sequence (a_n) to obtain sufficient cancellation.

(a) If $k \geq q$ and $f(k+1) \leq N$, we have perfect cancellation, i.e. $S_k = 0$. Indeed, in this case,

$$S_k = \sum_{n=f(k)}^{f(k+1)-1} e^{2\pi i n p/q} = \frac{\left(e^{2\pi i p/q}\right)^{f(k+1)} - \left(e^{2\pi i p/q}\right)^{f(k)}}{e^{2\pi i p/q} - 1}$$

by the geometric series formula. Since $k \geq q$, $(k+1)!$ is divisible by q, so the numerator vanishes:

$$\left(e^{2\pi i p/q}\right)^{f(k)+(k+1)!} - \left(e^{2\pi i p/q}\right)^{f(k)} = \left(e^{2\pi i p/q}\right)^{f(k)} \left(e^{2\pi i p(k+1)!/q} - 1\right) = 0.$$

(b) If $k \geq q$ and $N \in I_k$ (this happens for exactly one value of k), we have

$$S_k = \sum_{n=f(k)}^{N} e^{2\pi i n p/q} = \frac{\left(e^{2\pi i p/q}\right)^{N} - \left(e^{2\pi i p/q}\right)^{f(k)}}{e^{2\pi i p/q} - 1}.$$

By the triangle inequality, we deduce the bound

$$|S_k| \leq \frac{\left|\left(e^{2\pi ip/q}\right)^N\right| + \left|\left(e^{2\pi ip/q}\right)^{f(k)}\right|}{\left|e^{2\pi ip/q} - 1\right|} = \frac{2}{\left|e^{2\pi ip/q} - 1\right|}$$

which is independent of N.

(c) If $k \geq q$ and $N < f(k)$, we have of course $S_k = 0$ since the sum is empty.

Therefore,

$$\left|\sum_{n=1}^{N} a_n e^{2\pi in\theta}\right| \leq \sum_{k=0}^{q-1} |S_k| + \sum_{k\geq q} |S_k| \leq f(q) + \frac{2}{\left|e^{2\pi ip/q} - 1\right|}.$$

This shows that

$$\sup_{N\in\mathbb{N}} \left|\sum_{n=1}^{N} a_n e^{2\pi in\theta}\right| < +\infty.$$

10 Program Design For The Opening Ceremony

Suppose you are chosen as a technology assistant by the director of the opening ceremony for the 2022 Winter Olympics, and your job is to evaluate the program proposals. One of the backup programs is a skating show of a ensemble of drones dressed as mascots, which are moving along a circle. Since the number of the drones is sufficiently large, we can use a probability density function $\rho(t, v)(\geq 0)$ to represent the distribution of the drones. Here, $v \in \mathbb{R}$ is the line speed, and for a given time t, and two speeds $v_1 < v_2$,

$$\int_{v_1}^{v_2} \rho(t, v)\, dv$$

is the probability of find a drone with its speed between v_1 and v_2.

Suppose that the dynamics of the density function is governed by

$$\rho_t + \big((u(t) - v)\rho\big)_v = \rho_{vv}, \quad v \in \mathbb{R}, \quad t > 0,$$

where $u(t)$ is the command speed.

(1) To investigate proper choices of the command speed, D.B. propose that we shall choose

$$u(t) = u_0 + u_1 N(t)$$

where $u_0 > 0$, $u_1 > 0$ and $N(t)$ is the average of the positive part of the speed $v_+ = \max\{0, v\}$, i.e.,

$$N(t) = \int_{-\infty}^{+\infty} v_+ \rho(t, v)\, dv = \int_0^{+\infty} v\rho(t, v)\, dv.$$

But you claim that if $u_1 > 1$, $N(t)$ may become unbounded in evolution, such that the drones may be out of order. Can you prove it? (For simplicity, the contributions of ρ and its derivatives at $|v| \to +\infty$ are neglected.)

(2) After taking those advices, the directly is wondering whether the drones will be evenly distributed along the circle. Thus, we shall consider the joint density function $p(t, x, v)(\geq 0)$ of position and speed, where $x \in [0, 2\pi]$ is the position coordinate on the circle. Clearly, $\int_0^{2\pi} p(t, x, v)dx = \rho(t, v)$. Suppose that the governing equation for $p(t, x, v)$ is

$$p_t + vp_x + \big((u(t) - v)p\big)_v = p_{vv}, \quad x \in [0, 2\pi], \quad v \in \mathbb{R}, \quad t > 0.$$

Because, the drones are circulating around, the following boundary condition is satisfied

$$p(t, 0, v) = p(t, 2\pi, v), \quad v \in \mathbb{R}, \quad t > 0.$$

You have a feeling that, no matter how the drones are distributed initially, they will become almost evenly distributed very quickly. Can you prove or disprove this statement? (For simplicity, the contributions of p and its derivatives at $|v| \to +\infty$ are neglected.)

10 Answer

(1) We define the average speed

$$M(t) = \int_{-\infty}^{+\infty} v\rho(t,v)\,dv.$$

By direct calculations, we find

$$\begin{aligned}
\frac{d}{dt}M(t) &= \frac{d}{dt}\int_{-\infty}^{+\infty} v\rho(t,v)\,dv \\
&= \int_{-\infty}^{+\infty} v\rho_t(t,v)\,dv \\
&= \int_{-\infty}^{+\infty} v\left(-(u(t)-v)\rho+\rho_v\right)_v\,dv \\
&= \int_{-\infty}^{+\infty} \left((u(t)-v)\rho-\rho_v\right)\,dv \\
&= u(t) - M(t) \\
&= u_0 + u_1 N(t) - M(t).
\end{aligned}$$

Because $M(t) \le N(t)$ and clearly $N(t) \ge 0$, we get

$$\frac{d}{dt}M(t) \ge u_0 + (u_1 - 1)N(t) \ge u_0 > 0.$$

Then we conclude that when $t \to +\infty$, $M(t) \to +\infty$.

And since $N(t) \ge M(t)$, $N(t)$ also diverges to $+\infty$.

(2) Due to the periodic boundary condition in x, we can write the solution in the form of Fourier series:

$$p(t,x,v) = \frac{1}{2\pi}\sum_{k=-\infty}^{+\infty} p_k(t,v)e^{ikx}.$$

where

$$p_k(t,v) = \int_0^{2\pi} p(t,x,v)e^{-ikx}dx.$$

And we aim to show for $k \neq 0$, $p_k(t,v)$ shall decay.

Due to the orthogonality, we find p_k satisfies

$$\partial_t p_k + ikvp_k = -\partial_v\big((u(t)-v)p_k\big)_v + \partial_{vv}p_k, \quad t > 0, \quad k \in \mathbb{Z}, \quad v \in \mathbb{R}.$$

Next we apply the Fourier transform in v. Let

$$\hat{p}_k(t,\xi) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} p_k(t,v)e^{-iv\xi}dv,$$

and we can derive that $\hat{p}_k(t,\xi)$ satisfies

$$\partial_t\hat{p}_k + (\xi - k)\partial_\xi\hat{p}_k = -\left(i\xi u(t) + \xi^2\right)\hat{p}_k.$$

Next we use the method of characteristics. Consider the characteristic equation

$$\frac{d}{dt}\xi_k(t) = \xi_k(t) - k,$$

whose solution is given by

$$\xi_k(t) - k = e^{t-s}\left(\xi_k(s) - k\right).$$

Along the characteristics, we have

$$\hat{p}_k(t, \xi_k(t)) = \hat{p}_k(0, \xi_k(0)) e^{-\int_0^t (\xi_k(s))^2 + iu(s)\xi_k(s) ds}$$

which leads to

$$\hat{p}_k(t, \xi) = \hat{p}_k(0, (\xi - k)e^{-t} + k) e^{-H_{t,k}(\xi - k)},$$

where

$$H_{t,k}(z) = \frac{1 - e^{-2t}}{2} z^2 + 2k(1 - e^{-t})z + i\int_0^t u(s)e^{-(t-s)} ds z + k^2 t + ik\int_0^t u(s) ds.$$

Applying the inverse Fourier transform, we get

$$p_k(t, v) = e^{ikv}(p_{t,k} * G_{t,k})(v).$$

where

$$p_{t,k}(y) := e^t p_{0,k}(e^t y), \quad p_{0,k}(v) := e^{-ikv}\int_0^{2\pi} p_{\text{init}}(x, v) e^{-ikx} dx.$$

and $G_{t,k}(z)$ is a complex-valued Gaussian function defined as follows

$$G_{t,k}(z) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(z-\mu)^2}{2\sigma^2}\right) \exp\left(ik\Theta\right) \exp\left(-k^2 D\right).$$

$$\mu(t) = \int_0^t e^{-(t-s)} u(s) ds, \quad \sigma(t) = \sqrt{1 - e^{-2t}},$$

$$\Theta(t, z) = -\frac{2(z - \mu(t))}{1 - e^{-t}} - \int_0^t u(s) ds, \quad D(t) = t - 2\frac{1 - e^{-t}}{1 + e^{-t}}.$$

We observe that when $t > 0$, $G_{t,k}$ decays due to the factor $\exp\left(-k^2 D\right)$, and for t sufficiently large, it decays exponentially fast.

At last, with the convolution inequality, we get

$$\begin{aligned} \|p_k(t, v)\|_{L^1(\mathbb{R})} = \|e^{ikv}(p_{t,k} * G_{t,k})(v)\|_{L^1(\mathbb{R})} &\le \|p_{t,k}(g)\|_{L^1(\mathbb{R})} \|G_{t,k}\|_{L^1(\mathbb{R})} \\ &= \|p_{0,k}(g)\|_{L^1(\mathbb{R})} e^{-k^2 D(t)}. \end{aligned}$$

This shows that for $k \neq 0$, $p_k(t, v)$ decays, and $p(t, x, v)$ is approaching $\frac{1}{\pi} p_0(t, v)$, which is an even distribution in space.